

MySQL Built-in Functions

MySQL provides many built-in functions to perform calculations or format data directly inside your **SELECT** queries. They are divided into four main categories:

1. String Functions (Text manipulation)

Used to change or analyze text data.

- `UPPER(column_name)`: Converts text to UPPERCASE.
- `LOWER(column_name)`: Converts text to lowercase.
- `LENGTH(column_name)`: Returns the number of characters in a string.
- `CONCAT(col1, ' ', col2)`: Joins two or more strings together.

2. Math Functions (Number manipulation)

Used to perform mathematical operations.

- `ROUND(column_name, decimals)`: Rounds a number to a specified number of decimal places.
- `CEIL(column_name)`: Rounds a number UP to the nearest whole number (e.g., 4.2 becomes 5).
- `FLOOR(column_name)`: Rounds a number DOWN to the nearest whole number (e.g., 4.8 becomes 4).

3. Date & Time Functions

Used to work with dates and times.

- `NOW()`: Returns the current date and time.
- `CURDATE()`: Returns only the current date.
- `YEAR(date_column)`: Extracts only the year from a date.
- `DATEDIFF(date1, date2)`: `

4. Aggregate Functions (Summarizing data)

Used to perform a calculation on multiple rows and return a single summary value.

- `COUNT(column_name)`: Counts the number of rows.
- `SUM(column_name)`: Calculates the total sum of a numeric column.
- `AVG(column_name)`: Calculates the average value.
- `MAX(column_name)`: Finds the highest value.
- `MIN(column_name)`: Finds the lowest value.
-

Example : Employee Database

The Scenario / Questions: We have a table named `Staff` that stores employee details. First, create the table and insert the data below:

```
CREATE DATABASE OfficeDB;
USE OfficeDB;

CREATE TABLE Staff (
    EmpID INT AUTO_INCREMENT PRIMARY KEY,
```

```
FirstName VARCHAR(50),
LastName VARCHAR(50),
Salary DECIMAL(10,2),
HireDate DATE
);
```

```
INSERT INTO Staff (FirstName, LastName, Salary, HireDate) VALUES
('Kamal', 'Perera', 45500.50, '2020-05-15'),
('Nimali', 'Silva', 52300.75, '2019-11-20'),
('Sunil', 'Fernando', 38900.20, '2022-01-10'),
('Amala', 'Kumari', 61000.90, '2018-08-05');
```

Write SQL queries to answer the following using Built-in Functions:

1. **String:** Display the FirstName and LastName combined together as a single column named FullName. Make sure the full name is in **UPPERCASE**

```
select upper(concat(FirstName, ' ', LastName)) as FullName from staff;
```

2. **String:** Find out how many letters are in each employee's FirstName.

```
select FirstName, length(FirstName) as NameLenght from staff;
```

3. **Math:** Display the Salary of all employees **rounded** to the nearest whole number (no decimal places).

```
select FirstName, round(salary) as RoundedSalary from staff;
```

4. **Date:** Write a query to find out the **Year** each employee was hired.

```
select FirstName, year(HireDate) as HireYear from staff;
```

5. **Date:** Calculate how many **days** have passed from their HireDate to today's date (CURDATE()).

```
select firstName, datediff(curdate(), HireDate) as DaysOFWORKS from staff;
```

6. **Aggregate:** Find the **Total Salary** paid to all employees, the **Average Salary**, and the **Maximum Salary**.

```
select
    sum(salary) as TotalSalary,
    avg(salary) as TotalAvg,
    max(salary) as HighestSalary
from staff;
```